

SAFEATH AI: AN AI-DRIVEN SYSTEM FOR PERSONAL SAFETY

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INTRODUCTION

- Personal safety has become a major concern, especially for women and individuals travelling alone in public spaces.
- Existing safety applications mainly provide reactive support and lack intelligent real-time danger assessment.
- Delays in emergency response and lack of reliable reporting systems reduce the effectiveness of current safety solutions.
- SafePath AI is an integrated platform for risk detection, dangerous spot reporting, and emergency assistance.
- The system includes real-time safety prediction, SOS alerts, and community-based safety support.

PROBLEM STATEMENT

SafePath AI addresses several limitations in existing personal safety systems.

- Existing safety applications mainly provide reactive support after incidents occur.
- Lack of real-time danger prediction and intelligent emergency assistance.
- Delays in emergency response reduce the effectiveness of current systems.
- No unified platform combines risk prediction, SOS alerts, reporting, and journey safety.
- Limited accountability and verification in user-generated incident reports.

OBJECTIVES

01 Real-Time Risk Assessment

Predict safety score for user's current GPS location using Machine Learning.

02 Multi-Factor SOS

Phone shake and manual SOS button instantly trigger emergency alerts and location sharing.

03 Dangerous Spot Mapping

Users and police can report, verify, and visualize dangerous or safe locations.

04 Blockchain Credit Scoring

Ethereum smart contracts maintain transparent and tamper-proof user credit scores.

OBJECTIVES

05 Police Management Portal

Police dashboard to monitor SOS requests, verify reports, and manage emergency response.

07 Community Safety Network

Users can connect with journey creators, communicate through community chat, and travel together safely.

06 Journey Safety & Sharing

Users can share journeys, routes, and travel information with other users.

08 False Report Management

Police verify submitted reports, and users repeatedly reporting false incidents can be identified and blocked by the admin.

LITERATURE SURVEY

TITLE	AUTHOR	DESCRIPTION
Safe Path – A Real Time Women Safety Application (2023)	Dr. K. P. N. V. Satya Sree et al.	Uses GPS tracking, route suggestions, and SOS alerts for women safety assistance.
Herd Routes – A Pre-emptive IoT-Based System for Female Pedestrian Safety (2022)	Madeleine Woodburn et al.	Integrates IoT sensors and blockchain incentives for secure pedestrian safety reporting.
SafePath – Identifying Safe Risky Zones (2025)	Pratiksha P. Tidmesh et al.	Uses machine learning and crime data analysis to identify risky locations in real time.
WSA – A Navigation App for Women Safety (2023)	R. Jayabhaduri et al.	Provides safety index calculation, route visualization, and emergency alert features.
Achieving Anonymous and Covert Reporting on Public Blockchain Networks (2023)	Liehuang Zhu et al.	Uses blockchain-based reporting mechanisms to improve transparency and trust in public safety systems.

RESEARCH GAP IDENTIFIED

- Most systems rely on manual reporting and basic SOS mechanisms.
- Real-time AI-based safety prediction using live GPS context is limited.
- Blockchain-based verification and accountability are absent in existing platforms.
- No unified system combines risk prediction, SOS alerts, reporting, and journey sharing.
- Police coordination and centralized emergency monitoring remain limited.
- Community-based journey safety features are largely unavailable.

Overview

SafePath AI is an integrated personal safety platform that combines real-time risk prediction, emergency response, blockchain verification, and community-based safety support.

- **Real-time data collection** using GPS, accelerometer, and environmental APIs
- **AI-based risk prediction** using Machine Learning algorithms
- **Blockchain verification layer** for secure report validation and credit scoring
- **Django backend integration** for data processing and API management
- **Flutter mobile application** for SOS alerts, spot reporting, and journey sharing

SYSTEM ARCHITECTURE

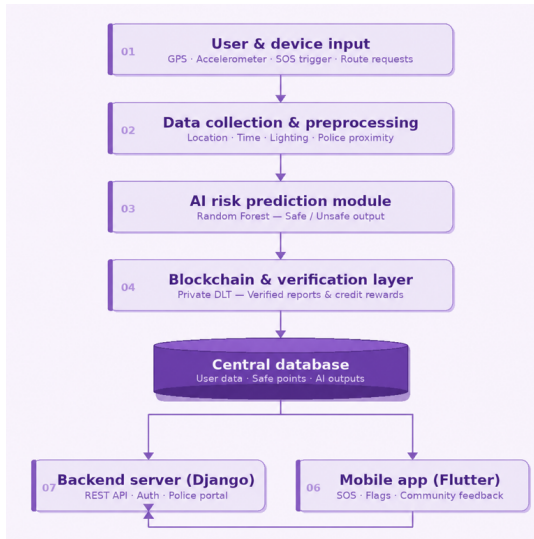


Figure 1: System Architecture of SafePath AI

DFD DIAGRAM



Figure: Data Flow Diagram (Level 0)

DFD DIAGRAM

Level 1

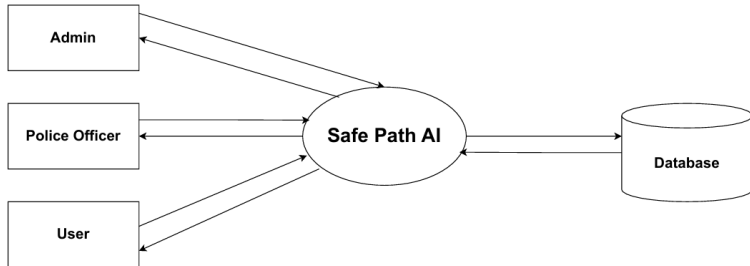


Figure: Data Flow Diagram (Level 1)

USE CASE DIAGRAM

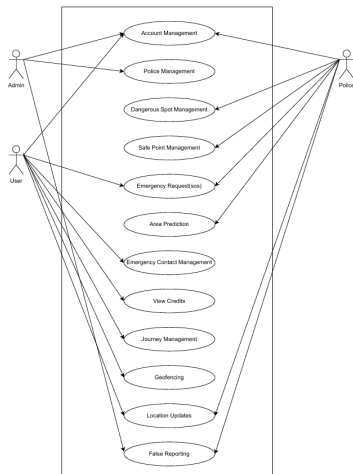


Figure: Use Case Diagram of the Proposed System

DATASET DESCRIPTION

Dataset 1: Chennai Pedestrian Safety Survey

A field survey dataset collected from women pedestrians in Chennai, India, containing environmental and contextual safety features.

- Real-world survey of 104 pedestrian locations
- 10 contextual safety features per record
- Safety labels: Safe, Unsafe
- Includes police station and bar proximity data
- Used as the primary dataset for model training

Source

<https://github.com/Srinithyee/ML-WOMEN-SAFETY>

DATASET DESCRIPTION

Dataset 2: Augmented Safety Dataset

Synthetic dataset generated using controlled randomisation to improve training size and model generalisation.

- 1,092 records generated from original 104 rows
- Same 10 features and safety labels retained
- Preserves real-world data distribution
- Used for Random Forest model training
- Achieved 95.43% prediction accuracy

Source

Generated as part of the SafePath AI Project

DATASET DESCRIPTION

Input Features Used for Training

Feature	Type	Example Values
Area	Categorical	Ramapuram, Adyar, West Mambalam
Zone	Categorical	Aanandam Nagar, Ambal Nagar
Time	Categorical	Morning, Afternoon, Evening, Night
People Frequency	Categorical	Low, Medium, High
Is Police Station	Binary	Yes, No
Is Bar	Binary	Yes, No
Tier	Categorical	Inner, Middle, Outer
Residence Level	Categorical	Low, Medium, High
Day of Week	Categorical	Monday – Sunday
Lighting	Categorical	Good, Moderate, Poor

Target Variable

Class: Safety classification label — *Safe* and *Unsafe*

SYSTEM REQUIREMENTS

Software Requirements

Python 3.x	Backend — ML, blockchain, Django
Django 3.0	Web framework, REST APIs, ORM
Flutter (Dart)	Cross-platform mobile app
scikit-learn	Random Forest prediction model
web3.py + Solidity	Ethereum blockchain via Ganache
Ganache	Local Ethereum blockchain tool
SQLite	Relational database
Nominatim + Overpass	GPS feature extraction (OSM)
PyCharm / Android Studio	IDEs

SYSTEM REQUIREMENTS

Hardware Requirements

Processor	Intel i5 / AMD Ryzen 5 or higher
RAM	Minimum 8 GB
Storage	SSD with adequate free space
Internet	Stable for Maps & SMS APIs
Smartphone	Android 6.0+ for Flutter app
OS	Windows 10 / Ubuntu 20.04

IMPLEMENTATION OVERVIEW

- Developed a multi-layer pedestrian safety platform using Flutter, Django, Machine Learning, and Blockchain
- Flutter mobile application handles GPS tracking, SOS alerts, reporting, and community interaction
- Django backend manages authentication, REST APIs, business logic, ML integration, and database operations
- Real-time location features extracted using Nominatim and Overpass APIs
- Random Forest machine learning model predicts safety levels from contextual environmental features
- Solidity smart contract on Ganache blockchain manages transparent user reward verification

FRONTEND IMPLEMENTATION

- Mobile app developed using **Flutter (Dart)**, tested on Android
- Admin and Police access the system via a **Django web portal**
- **Signup and Login** screens with role-based navigation
- **Safety Prediction page** — user taps predict, GPS is auto-fetched
- **Dangerous Spot Reporting** — user marks and submits location
- **Group Journey page** — post, browse, and request to join journeys
- **SOS screen** — triggered by shake gesture or manual button
- **Credit Score page** — displays user's blockchain score, verified spots, and level progress
- **Emergency Contacts** — manage contacts for SOS notification

BACKEND IMPLEMENTATION

- Backend built with **Django 3.0** and SQLite database
- 30+ URL routes for user, police, and admin modules
- ML model loaded at startup from `safety_model.pkl`
- Nominatim + Overpass extract **10 features** from GPS automatically
- **Web3.py** connects Django to Ganache Ethereum blockchain
- Solidity smart contract deployed at **fixed address** on local chain
- Police officer portal to **approve or reject** reported danger spots
- Flutter communicates with backend via **REST API (JSON)**

DATA ENGINEERING IMPLEMENTATION

- **Data Collection:** Real-world pedestrian safety survey of 104 locations in Chennai
- **Feature Selection:** 10 contextual features — area, zone, time, lighting, tier, residence level, police/bar proximity, people frequency, day of week
- **Data Augmentation:** Synthetic expansion from 104 to **1,092 records** using controlled randomisation
- **Label Encoding:** All categorical features encoded using **scikit-learn LabelEncoder**
- **Train-Test Split:** 80% training, 20% testing with **stratified sampling**
- **Model Persistence:** Trained model saved as **.pkl** files using **joblib**

MACHINE LEARNING ALGORITHM USED

- **Random Forest Classifier** used for safety level prediction
- Ensemble of **100 decision trees** with majority voting
- Each tree trained on a **random subset** of features and samples
- Handles mixed categorical and numerical data **without scaling**
- Resistant to overfitting via **max_depth = 10** constraint
- Outputs safety class: **Safe** and **Unsafe**
- Also returns **class probabilities** for confidence display
- Model achieves **95.43% accuracy** on held-out test set

MODEL TRAINING PROCESS

- Load augmented dataset (`augmented_safety_dataset.csv`)
- Encode all categorical columns using **LabelEncoder**
- Split into 80/20 train-test sets with **stratify=y**
- Train **RandomForestClassifier** with parameters:
 - `n_estimators`: 100
 - `max_depth`: 10
 - `random_state`: 42
 - `n_jobs`: -1 (all CPU cores)
- Evaluate using **accuracy score** and classification report
- Save model and encoders as `.pkl` files via `joblib`

SAFETY PREDICTION PROCESS

When a **GPS** coordinate is received by the system:

- Live GPS coordinates sent from **Flutter app** to Django server
- System checks against **hardcoded known locations** using Haversine distance
- If matched: pre-defined features used **directly**
- If unmatched: **Nominatim** reverse-geocodes area and zone; **Overpass API** checks for nearby police stations and bars
- Time, lighting and day of week derived from **system clock**
- All 10 features encoded and passed to **Random Forest model**
- Prediction and confidence score **returned to Flutter app**

BLOCKCHAIN IMPLEMENTATION

- Ethereum smart contract written in **Solidity 0.8.0** — stores user credit scores on-chain
- Contract deployed on **Ganache** — local Ethereum blockchain for development
- **Web3.py** connects Django backend to the Ganache blockchain
- When police **approves a reported spot**, Django calls `updateCreditScore()` on the contract
- `getUserByUsername()` fetches user score — read-only, no gas cost
- All score updates are **tamper-proof** — once written to a block, cannot be altered

TESTING

Testing — Functional Testing

- User login on mobile app opens user dashboard
- Admin and police login on web app opens correct portal
- GPS prediction returns correct safety class and confidence
- Police officer approves/rejects reported spots from web
- Approved spot triggers credit score update on blockchain
- SOS shake trigger activates alert and notifies contacts
- Group journey posted and visible to nearby users
- Dangerous spot report saved with correct coordinates
- Emergency contacts saved and retrieved per user

Testing — Integration Testing

- Mobile app and Django exchange data via REST API (JSON)
- GPS coordinates passed to ML module without errors
- Nominatim and Overpass correctly populate all 10 features
- ML prediction result returned and displayed on mobile
- Blockchain called from Django after police approval on web
- Web3.py transaction confirmed with valid receipt
- SOS alert created on mobile triggers end-to-end notification

Testing — Database Testing

- User and police records saved and retrieved correctly
- ForeignKey relationships maintained across all tables
- Spot status updates (pending → approved) reflect correctly
- SOSAlert stores GPS coordinates and timestamp accurately
- Chat messages retrieved in correct sender/receiver order
- All 12 migrations applied with no schema conflicts

Testing — ML & Blockchain Testing

- Random Forest achieves **95.43% accuracy** on test set
- Known locations matched correctly within radius threshold
- Nominatim and Overpass return correct features for test GPS
- Ganache connection established at `HTTP://127.0.0.1:7545`
- Smart contract deployed and address saved in `deployed.json`
- `updateCreditScore()` confirmed with valid block and gas
- `getUserByUsername()` returns correct score from chain

RESULT

This section presents the experimental results obtained from the implementation of the SafePath AI urban safety prediction system using a Random Forest classification model.

The evaluation focuses on measuring the performance of the prediction model using several metrics:

- Confusion Matrix Analysis
- Precision–Recall Evaluation
- F1-Score Analysis

These metrics demonstrate the effectiveness of the proposed system in accurately predicting safe and unsafe urban zones and assisting users in choosing safer travel routes.

Model Evaluation — Performance Metrics

- Model: **Random Forest Classifier** (100 trees, max_depth = 10)
- Training set: **873 samples** (80%) — Test set: **219 samples** (20%)
- Overall Accuracy: **95.43%**

Class-wise Performance

Class	Precision	Recall	F1-Score
Safe	0.98	0.94	0.96
Unsafe	0.93	0.97	0.95
Macro Avg	0.95	0.96	0.95
Weighted Avg	0.96	0.95	0.95

Model Evaluation — Confusion Matrix

- Test set contains **125 Safe** and **94 Unsafe** samples

Confusion Matrix

		Predicted	
		Safe	Unsafe
Actual	Safe	118	7
	Unsafe	3	91

Interpretation

- **118** Safe locations correctly identified — **7** misclassified as Unsafe
- **91** Unsafe locations correctly identified — **3** misclassified as Safe
- Only **3 false Positive** — system rarely marks a dangerous spot as safe
- High recall for Unsafe class (0.97) ensures strong **safety-critical reliability**

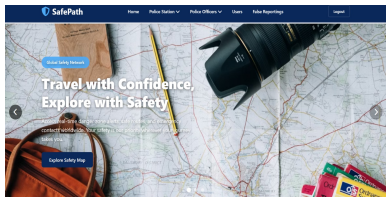
FUTURE SCOPE

- Expand dataset to cover regions beyond Chennai for better generalisation
- Integrate Google Maps and Places API for more accurate feature extraction
- Deploy smart contract on a public Ethereum testnet such as Sepolia
- Integrate real-time crime data from government APIs for live updates
- Add iOS support and publish mobile app to Play Store
- Implement route safety scoring — predict safety along an entire path
- Add multilingual support for regional languages including Malayalam
- Use advanced ML models such as XGBoost or deep learning for higher accuracy
- Enable real-time location sharing during active journeys for group safety

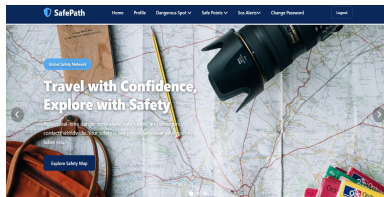
Conclusion

- Urban safety requires proactive and reliable solutions.
- The proposed SafePath AI system predicts safe and unsafe zones with **95.43% accuracy** using a Random Forest classifier.
- GPS-based automatic feature extraction eliminates manual input, enabling **zero-friction real-time predictions**.
- Users can report dangerous locations and receive safety alerts through the **Flutter mobile app**.
- Blockchain integration ensures **secure and tamper-proof** recording of user credit scores and contributions.
- Group travel matching and SOS alerts enhance safety through **community participation**.

SNAPSHOTS



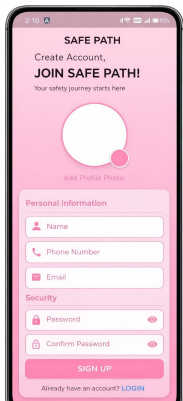
Admin Dashboard



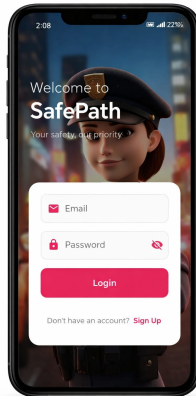
Police Dashboard

Web Portal — Admin & Police Dashboards

SNAPSHOTS

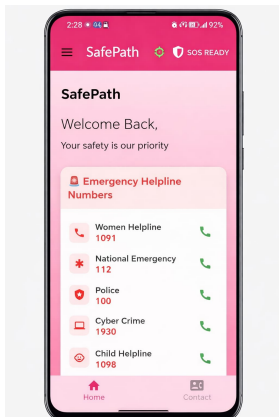


User Signup Page

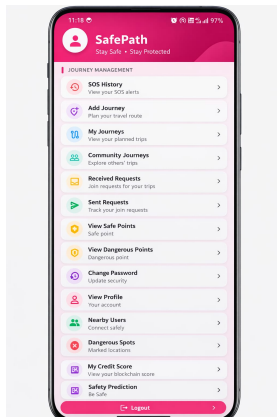


User Login Page

SNAPSHOTS

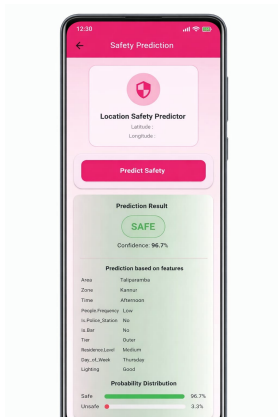


App Landing Page

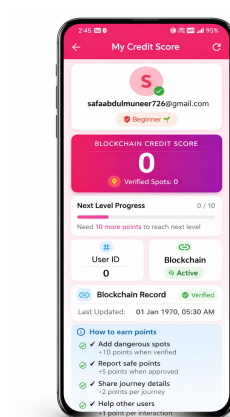


App Navigation Menu

SNAPSHOTS



Safety Prediction



Blockchain Credit Score

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Thank You!!